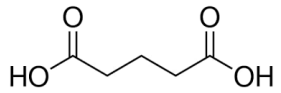
4(a)(i)	**M1** (molecules with the) same molecular formula / same number of atoms of each element **M2** different structural formulae/structures	**2**
4(a)(iii)	electrophilic addition	**1**
4(a)(iv)	**M1** CH_3COCH_3 + CH_3COOH as products **M2** $(CH_3)_2C = CHCH_3 + 3[O] \rightarrow CH_3COCH_3 + CH_3COOH$	**2**
4(a)(v)	**M1** (reaction proceeds via a) more stable intermediate / carbocation **M2** because of greater positive inductive effect of more alkyl groups	**2**

Examiner report missing
9701_w25_qp_24 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(c)(i)	homolytic	**1**
3(c)(ii)	**M1** $C_5H_{10} + Cl\bullet \rightarrow C_5H_9\bullet + HCl$ **M2** $C_5H_9\bullet + Cl_2 \rightarrow C_5H_9Cl + Cl\bullet$	**2**
3(c)(iii)	termination	**1**
3(d)(i)	HCl (g)	**1**
3(e)(i)		**1**
3(e)(ii)	**M1** absorption labelled at approx. 3000 cm^{-1} identified as O—H **M2** absorption labelled at approx. 1720 cm^{-1} identified as C=O	**2**

Examiner report missing
9701_w25_qp_23 Question 3

4(a)(i)	reduction	**1**
4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / $NaCl$ **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**

Examiner report missing
9701_w25_qp_23 Question 4

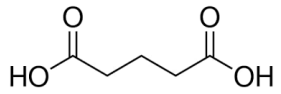
3(a)	**M1** (family of molecules / compounds / series with) same general formula **OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$. **M2** similar chemical properties / reactions	**2**
3(b)	**M1** strong C—H (bonds) **M2** non-polar (C—H / molecule / bonds)	**2**
3(c)	cracking	**1**
3(e)(i)	H_2 ((g)) **OR** hydrogen (gas)	**1**
3(e)(iii)	$C_6H_{14} + 9½O_2 \rightarrow 6CO_2 + 7H_2O$	**1**

Examiner report missing
9701_w25_qp_22 Question 3

4(a)(i)	free-radical substitution	**1**

4(a)(ii)	**M1** $C_3H_8 + Cl^\bullet \rightarrow CH_3CH_2CH_2^\bullet + HCl$ **OR** $C_3H_8 + Cl^\bullet \rightarrow C_3H_7^\bullet + HCl$ **M2** $CH_3CH_2CH_2^\bullet + Cl_2 \rightarrow CH_3CH_2CH_2Cl + Cl^\bullet$ **OR** $C_3H_7^\bullet + Cl_2 \rightarrow C_3H_7Cl + Cl^\bullet$ etc **M3** $CH_3CH_2CH_2^\bullet + Cl^\bullet \rightarrow CH_3CH_2CH_2Cl$ **OR** $C_3H_7^\bullet + Cl^\bullet \rightarrow CH_3CH_2CH_2Cl$	**3**

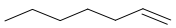
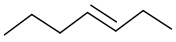
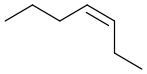
Examiner report missing
9701_w25_qp_22 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(c)(i)	homolytic	**1**
3(c)(ii)	**M1** $C_5H_{10} + Cl\bullet \rightarrow C_5H_9\bullet + HCl$ **M2** $C_5H_9\bullet + Cl_2 \rightarrow C_5H_9Cl + Cl\bullet$	**2**
3(c)(iii)	termination	**1**
3(d)(i)	HCl (g)	**1**
3(e)(i)		**1**

Examiner report missing
9701_w25_qp_21 Question 3

4(a)(i)	reduction	**1**
4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / NaCl **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**

Examiner report missing
9701_w25_qp_21 Question 4

5(a)	**M1**  **M2**  **OR** 	**2**
5(d)(i)	oxidising agent	**1**
5(d)(ii)	hot concentrated acidified $KMnO_4$	**1**

Examiner report missing
9701_s25_qp_24 Question 5

3(a)(i)	Al_2O_3 / aluminium oxide	**1**

3(a)(ii)	• unburned hydrocarbons • react with NO / NO_2 • (to) produce PAN (– a component of photochemical smog) Any 2 points = 1 mark, all 3 points = 2 marks	**2**

3(b)(i)	nickel / platinum	**1**

Examiner report missing
9701_s25_qp_23 Question 3

5(a)(i)	C_2H_3O	**1**

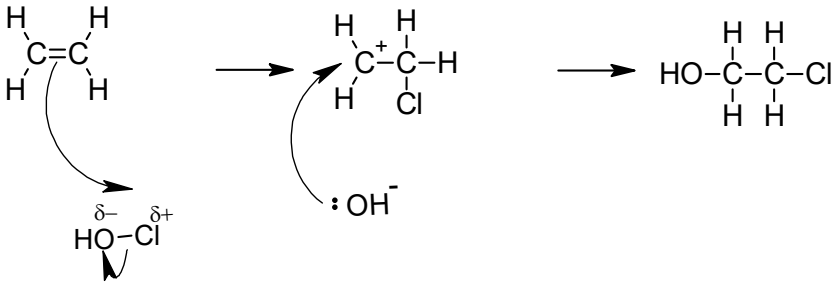
5(a)(ii)	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	**4**
	Na	fizz / bubbles etc / **OR** solid / Na disappears / dissolves	$CH_2C(CH_3)CO_2Na$ **OR** $CH_2C(CH_3)CO_2^-Na^+$	
	Br_2 in the dark	brown to colourless / decolourise	$CH_2BrCBr(CH_3)CO_2H$	

5(c)	ethene pi bond is a region of (high) electron density (above and below the C-C (sigma) bond)	**1**

Examiner report missing
9701_s25_qp_23 Question 5

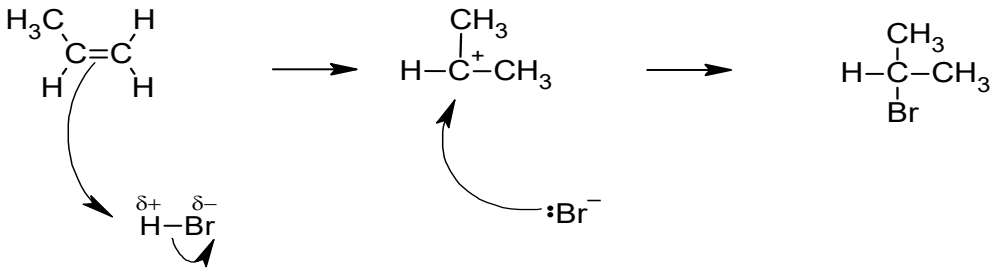
6(a)(i)	two or more reactants / molecules react to make (only) one product	**1**

6(a)(iii)	water purification / killing bacteria or microbes	**1**

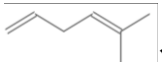
6(b)	 **M1** correct arrow for HO-Cl **M2** arrow from double bond of C=C to Cl **M3** correct intermediate **M4** arrow from lone pair on O of :OH- to the appropriate C	**4**

Examiner report missing
9701_s25_qp_23 Question 6

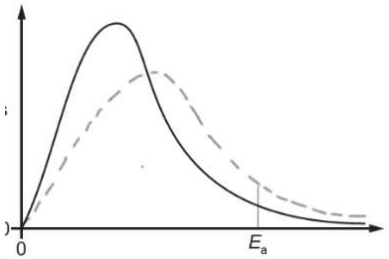
3(a)(i)	two (or more) reactants / molecules react to make (only) one product	**1**

3(a)(ii)	 **M1** curly arrow from = of C=C to H (of HBr) **M2** curly arrow from bond of $H^{\delta+}$—$Br^{\delta-}$ to $Br^{\delta-}$ (dipole is essential) **M3** correct intermediate **M4** curly arrow from lone pair of Br^- to C^+ of intermediate drawn	**4**

3(a)(iii)	**M1**(2-bromopropane is made from the) more stable intermediate / carbocation **M2** (due to positive) inductive effect of more alkyl groups (attached to the C^+ / central carbon atom) **OR** (due to) greater (positive) inductive effect of 2 methyl groups (rather than 1 ethyl group)	**2**

3(e)	$H_2C=CHCH_2CH=C(CH_3)_2$ ✓✓  ✓✓ $H_2CCHCH_2CHC(CH_3)_2$ ✓✓ **M1** $H_2C=C$…. on one end of their molecule **OR** …$C=C(CH_3)_2$ on one end of their molecule **OR** …$C=CHCH_2CH=C$… in the middle of their molecule **M2** correct structure	**2**

Examiner report missing
9701_s25_qp_22 Question 3

3(a)(i)	 *Two correct for one mark, all four for two marks:* • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only **AND** to the right of the original peak	**2**

3(a)(iii)	catalyst	**1**

Examiner report missing
9701_s25_qp_21 Question 3

4(a)	$C_3H_4O_3$	**1**
4(b)(i)	$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5}$ mol	**1**
4(b)(iv)	**M1** (also) oxidises / reacts with (primary and / or secondary) alcohol groups **M2** (also) breaks the sigma and pi bond in C=C	**2**

Examiner report missing
9701_s25_qp_21 Question 4

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	**1**
1(c)(ii)	ethanol (solvent) **AND** heat	**1**

Examiner report missing
9701_m25_qp_22 Question 1

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	**1**
3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) **AND** permanent dipole—permanent dipole / pd—pd (forces)	**1**
3(d)(ii)	electrophilic addition	**1**
3(d)(iii)		**2**
3(d)(v)	• secondary (carbo)(cat)ion/intermediate is more stable / $[CH_3CHCH_2I]^+$ is preferred • electrophile approaches with I nearest / $I^{\delta+}$ / I^+ attacks first • I and Br have different electronegativities / IBr is a polar molecule	**2**

Examiner report missing
9701_m25_qp_22 Question 3

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(d)(i)	**M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	**1**
	M2 $x = 3$	**1**

1(d)(iii)	a **single** product is made	**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

(b) Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

(c) (i) The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

(ii) Many candidates incorrectly used all three isotopes to calculate the A_r of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

(d) (i) Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of H_2O to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

(ii) Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

(iii) Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra CH_2 units to the chain.

iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(ii)	oxidising strength **decreases** from chlorine to iodine	**1**
2(a)(iii)	green colour disappears	**1**
2(a)(iv)	H—Hal covalent bond strength **decreases** down the group	**1**
2(b)(i)	a species with one or more unpaired electrons	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(b)(iii)		**2**

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

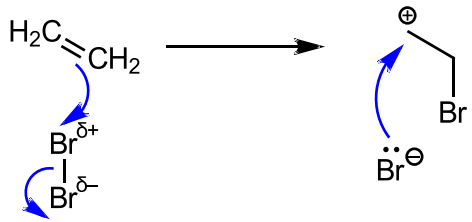
(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(d)(i)	**M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	**1**
	M2 $x = 3$	**1**

1(d)(iii)	a **single** product is made	**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

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iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(iii)	green colour disappears	**1**
2(b)(ii)	ultraviolet	**1**
2(b)(iii)	$C_2H_6 + Cl^{\bullet} \rightarrow C_2H_5^{\bullet} + HCl$	**1**
	$C_2H_5^{\bullet} + Cl_2 \rightarrow C_2H_5Cl + Cl^{\bullet}$	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

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(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

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5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(b)(i)	step 1 …..Br_2 (in the dark) step 2 …..NaOH(aq) + heat	**2**
5(b)(ii)	electrophilic addition	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

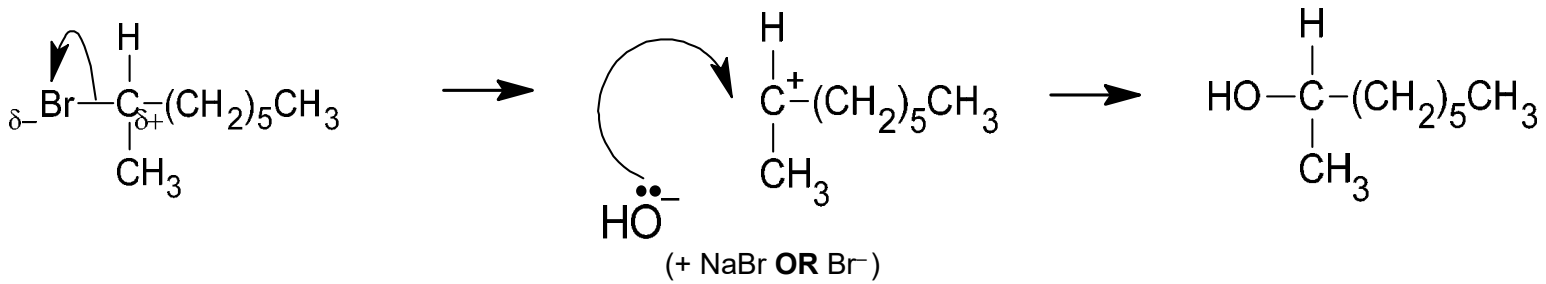
(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	Three-dimensional stereoisomers of $CH_3(CH_2)_5CHBrCH_3$ **M1** correct representation of left-hand answer as 3d structure of one of the enantiomers of $CH_3(CH_2)_5CHBrCH_3$ **M2** correct representation of 3d structure of the second enantiomer of $CH_3(CH_2)_5CHBrCH_3$	**2**
4(b)	 (+ NaBr **OR** Br^-) **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) **AND** curly arrow from bond of C—Br to Br (or just beyond) **M2** correct intermediate **ALLOW** R for $–(CH_2)_5CH_3$ **M3** curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where C–OH bond is made) *If $SN2$ mechanism shown:* Apply $SN2$ mechanism mark scheme **only** if mechanism shows OH / NaOH species attacking the reactant and no OH **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) AND curly arrow from bond of C—Br to Br of reactant (or just beyond) **NOT M2** **M3** curly arrow from lone pair on $:OH^-$ to the appropriate C of the reactant	**3**

4(c)	<table><tr><td>Chemical test</td><td>observation with 2-bromooctane</td><td>observation with octan-2-ol</td><td>observation with $CH_3(CH_2)_5CHCH_2$</td></tr><tr><td>Br_2 in the dark</td><td>×</td><td></td><td>orange / brown to colourless</td></tr><tr><td>PCl_5</td><td></td><td>steamy fumes</td><td>×</td></tr><tr><td>$AgNO_3(aq)$</td><td>cream precipitate **OR** off white precipitate</td><td>×</td><td></td></tr></table> 6 correct answers = 3 marks 4 or 5 correct answers = 2 marks 2 or 3 correct answers = 1 mark 0 or 1 correct answer = 0 marks	**3**
4(d)(i)	elimination	**1**
4(d)(ii)	NaOH / sodium hydroxide in ethanol	**1**

Question 4

(a) Some excellent representations of the three-dimensional structures were seen; these included the approximate bond angles for these tetrahedral structures. Other diagrams showed confusion over the types of symbols used to represent bonds in the plane with those in and out of the plane. In order to remove ambiguity, candidates should be encouraged to use the convention described in the syllabus, under expected conventions for representing organic structures.

(b) Good knowledge of the S_N1 mechanism was described with precision in the best answers. Some confusion regarding the relative charges on the C–Br dipole and either no reference to a lone pair of electrons on O of OH^- or the location of the lone pair of electrons on the H of OH^- were common incorrect details. Some answers confused the S_N2 mechanism details with S_N1.

(c) The observations of these reactions were generally well known. Some answers did not include observations, as directed in the table heading, rather than attempted to predict the product of the reactions.

(d) (i) The name of this reaction was quite well known. Addition, substitution, dehydration and cracking were common incorrect answers.

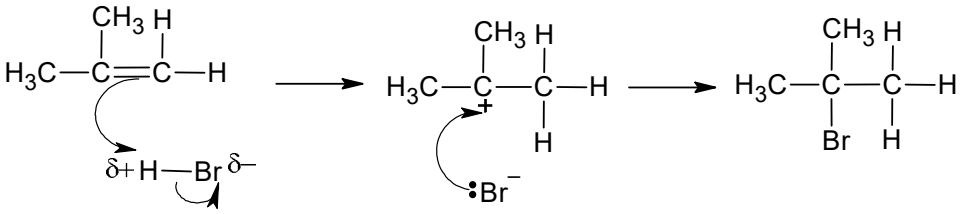
(ii) The reagent NaOH was described in many answers. Confusion of the relevant conditions was seen in weaker responses.

(e) (i) Fully correct answers were seen in around half of responses. Sometimes confusion between two of the types of hybridisations was seen. The weakest answers demonstrated a lack of understanding of the different types of hybridisations which occur in carbon atoms in order to produce single, double and triple covalent bonds.

(ii) It was common for answers to contain features which lead to both types of stereoisomerism rather than focus on the stereoisomerism present in unbranched hydrocarbons. A relatively small proportion of answers showed good understanding of the features of an alkene required in order for it to exist as geometrical (cis/trans) isomers. A lack of precision in answers produced some ambiguity in responses, for example ‘no movement of a double bond’ rather than ‘no rotation of a double bond’.

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	**1**

5(c)(i)	**EITHER** (pair of) electrons in π bond are further away from the nuclei so weaker attraction **OR** (pair of) electrons in σ bond are closer to the two nuclei so stronger attraction	**1**

5(c)(ii)	 **M1** arrow from = of C=C to H of H-Br **M2** correct dipole on H-Br **AND** arrow from H-Br bond to Br **M3** correct intermediate **AND** 2-bromopropane **M4** arrow from lone pair on Br^- to C^+ of their intermediate (with +ve or δ+ charge)	**4**

Question 5

(a) A minority of candidates gave the correct definition for a covalent bond. The most common errors included omitting a pair of electrons (being shared) and not mentioning that the attraction, for these electrons, was between the nuclei (of the two atoms).

(b) (i) The hybridisation of each carbon, in a C=C bond, was generally well answered.

(ii) Many candidates found this question difficult; some diagrams representing a σ bond were correct showing a linear overlap of two atomic orbitals or hybrids.

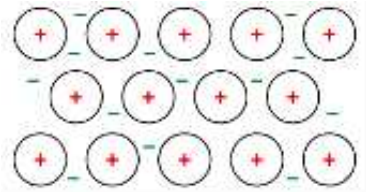
The diagrams for the formation of a π bond were often incorrectly drawn.
The most common mistakes included:

- not labelling p-orbitals, involved in the π bond formation
- drawing only the 'resultant' stage of π bond formation, showing the delocalised electrons above and below the nuclei, but without any labelling.

(c) (i) Credit was rarely awarded and most answers included a statement on how the σ and π bonds are formed by linear and sideways overlap of the relevant orbitals, which was then considered to render the π bond weaker than the σ bond.

The key idea is that during an electrophilic addition reaction the electrons in the π bond are involved as they are further from the carbon nuclei in the C=C bond and thus experience a weaker force of attraction from these nuclei.

(ii) Candidates who understood the reaction usually gained full credit. The main error involved the careless drawing of arrows, which often started from a C atom instead of from the middle of the π bond or from the Br, in a Br^- ion, instead of from a lone pair on the Br^- ion.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**

1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	**1**

1(d)(ii)	($KMnO_4$ is) **cold** and **dilute**	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

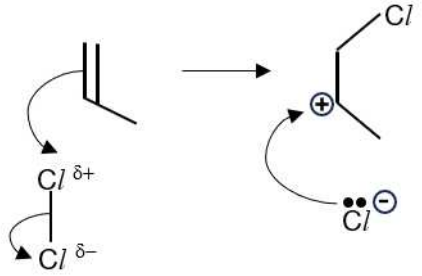
(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

2(a)	<table><tr><td>formula of chloride</td><td>● $NaCl$</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**

2(d)(i)	 ● curly arrow from C=C bond to $Cl^{(\delta+)}$	**1**
	● correct dipole on Cl_2 **AND** curly arrow from Cl–Cl bond to $Cl^{\delta-}$	**1**
	● correct carbocation intermediate	**1**
	● curly arrow from lone pair on Cl^- to C^+	**1**

2(d)(ii)	NaOH in ethanol **AND** heat	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

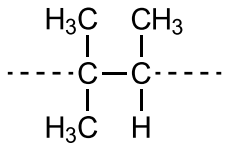
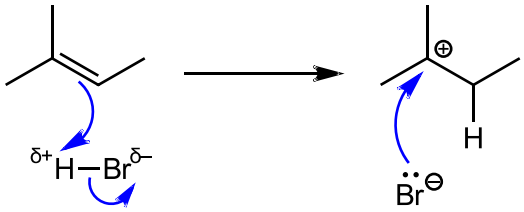
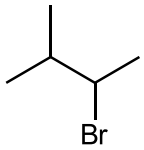
(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

4(a)(i)	3-methylbut-1-ene	**1**
4(a)(iv)	$(CH_3)_2CHCH_2CH_3$	**1**
4(a)(v)		**1**
4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**
4(c)(ii)		**1**

4(c)(iii)	**●C has a tertiary carbocation and a 2ndry carbocation** **●D has a secondary carbocation and a primary carbocation** **● alkyl groups have a +ve / positive / +I / inductive effect** **● more alkyl groups / more inductive effects give a <u>more stable</u> intermediate / carbocation / C+** ● **In C** tertiary carbocation is more stable than the secondary carbocation Any **two** ● for 1 mark Any **four** ● for 2 marks	**2**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified <u>concentrated</u> $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed <u>concentrated</u> H_2SO_4 (as a catalyst).

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(c)(i)	H_2O / steam	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

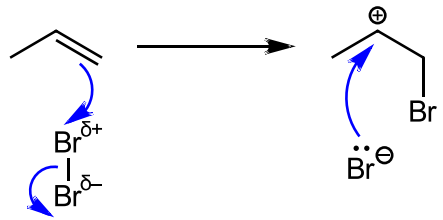
(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

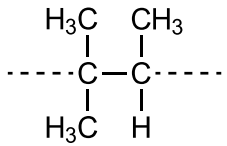
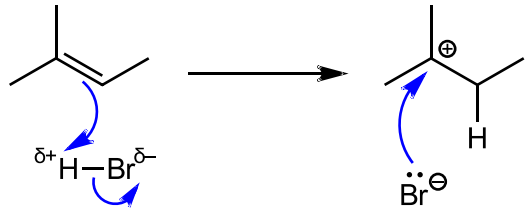
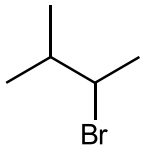
(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

4(a)(i)	3-methylbut-1-ene	**1**
4(a)(iv)	$(CH_3)_2CHCH_2CH_3$	**1**
4(a)(v)		**1**
4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**
4(c)(ii)		**1**

4(c)(iii)	**●C has a tertiary carbocation and a 2ndry carbocation** **●D has a secondary carbocation and a primary carbocation** **● alkyl groups have a +ve / positive / +I / inductive effect** **● more alkyl groups / more inductive effects give a <u>more stable</u> intermediate / carbocation / C+** ● **In C** tertiary carbocation is more stable than the secondary carbocation Any **two** ● for 1 mark Any **four** ● for 2 marks	**2**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

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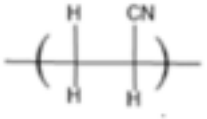
(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified concentrated $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

6(a)	 OR 	**1**
6(b)(i)	nucleophilic addition	**1**
6(b)(iii)	heat + conc. sulfuric acid / conc. phosphoric acid **OR** heat + Al_2O_3	**1**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

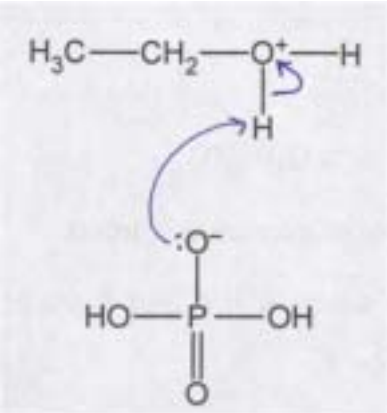
(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**
4(c)(i)	acid **AND** it donates a proton / H^+	**1**
4(c)(iii)	 **M1** arrow from bond of one of the O-H to O of the cation **M2** arrow from lone pair on O of the anion to between H of H-O	**2**

4(c)(vi)	**M1** (propan-2-ol is produced via a) more stable carbocation OR more stable intermediate **OR** $(CH_3)_2C^+H$ is more stable **M2** (positive) inductive effect of more alkyl/methyl groups **OR** more electron donating alkyl / methyl groups **OR** greater electron donating effect of two alkyl / two methyl groups	**2**

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

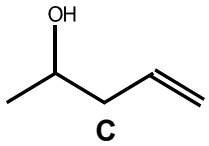
(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**

5(b)(i)	carbonyl (group)	**1**

5(c)(i)	 **C** **M1** $CH_3CH(OH)$ present **M2** five carbons in a straight chain **AND** molecular formula $C_5H_{10}O$ **AND** 1 C=C structure which shows only one type of stereoisomerism	**2**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

3(d)(i)	ultraviolet light	**1**

3(d)(ii)

name of stage	equation	
initiation	$Cl_2 \rightarrow 2Cl\bullet$	[1]
propagation	$CH_2Cl2 + Cl\bullet \rightarrow CHCl_2\bullet + HCl$	[1]
termination	$CHCl_2\bullet + Cl\bullet \rightarrow CHCl_3$	[1]

3

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

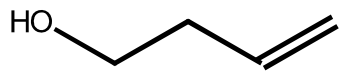
(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

4(a)(i)		**1**

4(b)(i)	cold dilute acidified $KMnO_4$	**1**

Question 4

(a) (i) Common errors were to retain the COOH group and to have the C=C double bond reduced.

(ii) This was well answered by most candidates.

(b) (i) This was generally well answered, although few candidates stated all the required conditions.

(ii) This was well answered by most candidates.

(iii) A small proportion of candidates gained full credit here.

(c) Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline I_2(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

5(a)(i)	$C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$ **M1** correct formula of M C_8H_{18} **M2** balanced equation showing complete combustion of M	**2**

5(a)(ii)	• C • CO • unburnt hydrocarbons • less/no CO_2 Two correct for one mark, three correct for two marks.	**2**

Question 5

(a) (i) This was well answered by most candidates, although balancing the equation proved challenging for some.

(ii) Most candidates correctly mentioned the production of CO instead of CO_2. A very small number correctly referred to C and unburnt hydrocarbons.

(b) (i) A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of N_2.

(ii) A small number of candidates correctly identified the catalytic role of NO_x and gave all the relevant equations. Many gave the equation for the formation of H_2SO_4, and a significant number gave equations and answers related to the formation of HNO_3 for acid rain.

(iii) This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

(c) (i) Candidates often omitted dilute/(aq) with a named acid here.

(ii) A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

(iii) This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

(iv) A small proportion of candidates answered this question correctly.

(v) A small proportion of candidates answered this question correctly.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(a)(iv)	96% × (5000 / 42.0) × 72.0 = 8230 or 8229 (kg)	**1**

3(b)(ii)	acidified $K_2Cr_2O_7$	**1**
	Tollens' reagent	**1**
	no reaction	**1**

3(c)	*alkane* C_3H_8	**1**
	alcohols $C_4H_{10}O$	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(a)(iii)	$C_4H_{10}O$ / $CH_3CH(OH)C_2H_5$ → C_4H_8 / $CH_3CHCHCH_3$ + H_2O	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.